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29 years old



Rémi Boutin

Education

- 2020–2023 **PhD, Mathématiques Appliquées à Paris 5, Université Paris Cité**, under the direction of Pr. Pierre Latouche and co-direction of Pr. Charles Bouveyron, Deep probabilistic models and inference strategies for the analysis of networks with textual-edges
- 2017–2020 **Engineer in statistics, ENSAE Paris**, Major: data science; minor: statistical learning
- 2016–2017 **Master 1 Université Paris 1 - Panthéon Sorbonne**, Major: applied mathematics; minors: economics, with highest honour
- 2013–2016 **Bachelor Université Paris 1 - Panthéon Sorbonne**, Applied mathematics to social sciences, with highest honour
- 2013 **Baccalaureate Sciences major, Highschool Descartes Antony**, speciality mathematics, with distinction

Languages French native language, English fluent

Computer science Python ● ● ● ● ● Pytorch ● ● ● ● ● Keras ● ● ● ● ● R ● ● ● ● ● C ● ● ● ● ●

Course contents

Majors, Statistical learning, Monte Carlo and sequential Monte Carlo, Bayesian statistic, optimisation, optimal transport, neural networks, measure and probability theory, econometrics, analysis, time series, algebra

Minors, Macroeconomics, microeconomics

Academic work

- 2024 - 2025 **Post-doctorat, Laboratoire de Probabilité, Statistique et Modélisation, Sorbonne Université**, under the supervision of Catherine Mathias, Sparse hypergraph modelling and application in ecology
- 2020–2023 **PhD, Mathématiques Appliquées à Paris 5, Université Paris Cité**, under the direction of Pr. Pierre Latouche and co-direction of Pr. Charles Bouveyron, Deep probabilistic models and inference strategies for the analysis of networks with textual-edges
- 2020 **Research internship**, Université Paris Cité, under Pr. Pierre Latouche's supervision, Graphical model and neural topic model for cluster detection in a network with textual edges
3 months
- 2019 **Research assistant**, Argonne National Laboratory, Chicago, USA, under Dominik Korbowski's supervision, Creation of a generative model of car speed profile based on the WaveNet architecture
6 months

Publications

- 2024 **The Deep Latent Position Block Model for the Clustering of Nodes in Multi-Graphs**, Seydina Ousmane Niang, Charles Bouveyron, Marco Corneli, Pierre Latouche, RB
- 2024 **The Deep Latent Position Block Model For The Block Clustering And Latent Representation Of Networks**, RB, Pierre Latouche and Charles Bouveyron, published in Statistics and Computing
- 2023 **The Deep Latent Position Topic Model for Clustering and Representation of Networks with Textual Edges**, RB, Pierre Latouche and Charles Bouveyron, published at Scandinavian Journal of Statistics
- 2023 **Embedded topics in the stochastic block model**, RB, Pierre Latouche and Charles Bouveyron, published in Statistics and Computing

Teaching

- 2024-2025 **Machine Learning 2 (40 H)**, *Polytech Clermont*, 4th year of mathematical engineering and data science, Key point : clustering, mixture models (Gaussian mixture model, EM algorithm), Bayesian modelling, introduction to Gaussian processes
- 2020-2022 **Seminal supervisor of Probability**, *Technological Institute of Université Paris Cité*, 1st year students
- 2019-2020 **Private lessons for bachelor students**, *2nd year of bachelor degrees in applied mathematics*, in analysis and algebra
- 2017-2018 **Mathematical tutor**, *Université Panthéon Sorbonne*, for students in 1st and 2nd year of bachelor's degree in applied mathematics

Conference/Seminar

- 2025 **Post-doc & Phd seminar**, *LMBP, Clermont-Ferrand*
A brief introduction to probabilistic modelling through the Gaussian mixture model and the EM algorithm.
- 2024 **Graphical models and Clustering**, *Conference, Montpellier, 2024*
The Deep Latent Position Topic Model for Clustering and Representation of Networks with Textual Edges.
- 2024 **Seminar of Statistic, CEREMAD**, *Université Paris-Dauphine PSL, Paris*
Deep Latent Position Block Model for Block Clustering and Latent Representation of Nodes in Networks.
- 2024 **55th Days of Statistic**, *Conference, Bordeaux, 2024*
The Deep Latent Position Block Model.
- 2024 **Seminar of Statistic, LPSM**, *Sorbonne Université, Paris*
The Deep Latent Position Topic Model for network with textual edges.
- 2023 **54th Days of Statistic**, *Conference, Brussels*
The Deep Latent Position Topic Model for Clustering and Representation of Networks with Textual Edges.
- 2023 **Statlearn**, *Conference & poster session, Montpellier*
The Deep Latent Position Topic Model for Clustering and Representation of Networks with Textual Edges.
- 2022 **Seminar of Probability, Analysis and Statistic, LMBP**, *Université Clermont Auvergne, Aubière*
The embedded topic for the stochastic block model.
- 2022 **Post-doc & Phd seminar, LMBP**, *Université Clermont Auvergne, Aubière*
Introduction to probabilistic modelling and variational inference.
- 2022 **Statlearn**, *Conference & poster session, Cargèse*
The embedded topic model meets the stochastic block model.
- 2021 **SophIA summit**, *Poster session, Sophia-Antipolis*
The embedded topic model meets the stochastic block model to cluster graph with textual data.

Work experiences

- 2018-2019 **Data Scientist internship**, *Saint-Gobain Paris, FRA*, under Nawaaz Peerbocus' supervision, 6 months
Identification of the causes of flaws in glass production
- 2017-2018 **Applied statistics project**, *ENSAE Paris*, under Claude Petit's supervision, Detection of important twitter accounts in the political landscape, based on a random walk and MCMC inference after *scrapping* Twitter
- 2017 **Hackathon ENSAE Paris and EMMAÜS Connect**, *Creation of a neural network classifier for automatic image annotation*

About my interests

- Arts** Movies (festival of short-movies), literature (classic, polar, SF)
- Sports** Fencing (regional referee), (mountain) bike
- Daily life** Climate (change), comedians